# Unit Circle — Degree-Radian and Quadrant-Sign Errors Answer Key

Precalculus

**Quick Answers**

1. $-\frac{1}{2}$
2. $\frac{1}{2}$
3. $-\frac{1}{2}$
4. $-\frac{\sqrt{3}}{3}$
5. $\frac{5\pi}{4}$
6. 210, 330
7. $\frac{\pi}{4}$, $\frac{7\pi}{4}$
8. -0.99

**Common wrong answers**

2. *If they got -0.99: evaluated 30 radians instead of 30 degrees (calculator left in radian mode)*
3. *If they got 0.5: used the reference angle's value without applying the quadrant III sign*
4. *If they got 0.58: kept the reference angle's positive tangent, ignoring that tangent is negative in quadrant II*
5. *If they got 12891.6: multiplied by 180 over pi instead of pi over 180, converting the wrong way*
8. *If they got 0.9986: read the 3 as 3 degrees instead of 3 radians*

## 1. Exact value of $\sin\frac{7\pi}{6}$.

Step 1: locate the angle. $\pi < \frac{7\pi}{6} < \frac{3\pi}{2}$, so the terminal side is in quadrant III. Step 2: reference angle: $\frac{7\pi}{6} - \pi = \frac{\pi}{6}$, whose sine is $\frac{1}{2}$. Step 3: apply the quadrant's sign. In quadrant III the $y$-coordinate is negative and sine reads the $y$-coordinate, so the value is negative. Step 4: $\sin\frac{7\pi}{6} = \boxed{-\frac{1}{2}}$.

## 2. Find and fix: Reva's $\sin 30° \approx -0.99$.

Step 1: name what the machine did. In radian mode `sin(30)` is the sine of 30 *radians*, an angle of nearly five full turns — not the sine of 30°. Step 2: notice the diagnostic sign. 30° is a quadrant-I angle, so its sine must be positive; a negative output is impossible and flags the mode before any value is looked up. Step 3: the exact value comes from the unit circle: $30° = \frac{\pi}{6}$ and $\sin\frac{\pi}{6} = \boxed{\frac{1}{2}}$. Step 4: to check on a calculator, either switch to degree mode or evaluate $\sin(\pi/6)$ in radian mode — both give 0.5.

## 3. Find and fix: Ari's $\cos\frac{4\pi}{3} = \frac{1}{2}$.

Step 1: Ari's reference angle is right: $\frac{4\pi}{3} - \pi = \frac{\pi}{3}$, and $\cos\frac{\pi}{3} = \frac{1}{2}$. Step 2: name the skipped step — the reference angle gives only the *size* of the value. The sign comes from the quadrant, and $\frac{4\pi}{3}$ lies in quadrant III ($\pi < \frac{4\pi}{3} < \frac{3\pi}{2}$). Step 3: in quadrant III the $x$-coordinate is negative, and cosine reads the $x$-coordinate, so the cosine is negative there. Step 4: $\cos\frac{4\pi}{3} = \boxed{-\frac{1}{2}}$.

**4. Find and fix: Sena's $\tan 150^\circ \approx 0.58$.**
Step 1: quadrant. $90^\circ < 150^\circ < 180^\circ$, so $150^\circ$ is in quadrant II. Step 2: sign. In quadrant II sine is positive and cosine is negative, and $\tan = \frac{\sin}{\cos}$, so the tangent must be *negative* there. Sena's positive answer is the tangent of the reference angle only. Step 3: reference angle: $180^\circ - 150^\circ = 30^\circ$, and $\tan 30^\circ = \frac{\sqrt{3}}{3}$. Step 4: attach the quadrant's sign: $\tan 150^\circ = \boxed{-\dfrac{\sqrt{3}}{3}}$, which is about $-0.58$.

**5. Find and fix: Bo's conversion of $225^\circ$ to radians.**
Step 1: size check first. A full turn is $2\pi \approx 6.28$ radians, so any angle under one turn converts to a radian measure below 6.28. Bo's 12891.6 is thousands of turns — the factor was applied upside down. Step 2: fix the factor. Degrees to radians multiplies by $\frac{\pi}{180}$ (radians to degrees is the reciprocal $\frac{180}{\pi}$, which is what Bo used). Step 3: convert: $225 \cdot \frac{\pi}{180} = \frac{225\pi}{180}$. Step 4: simplify by dividing both parts by 45: $\boxed{\dfrac{5\pi}{4}}$, which is about 3.93 radians — under $2\pi$, as the size check demanded.

**6. Solve $2\sin\theta + 1 = 0$ on $0^\circ \le \theta < 360^\circ$; diagnose Cal's $30^\circ$ and $150^\circ$.**
Step 1: isolate the ratio: $2\sin\theta = -1$, so $\sin\theta = -\frac{1}{2}$. Step 2: reference angle from the size of the ratio: $\sin 30^\circ = \frac{1}{2}$, so the reference angle is $30^\circ$ — this is the part Cal got right. Step 3: choose quadrants from the *sign*. The sine is negative in quadrants III and IV, not I and II; Cal placed his reference angle in the two quadrants where sine is positive, which is why both of his answers satisfy $\sin\theta = +\frac{1}{2}$ instead. Step 4: build the angles: quadrant III gives $180^\circ + 30^\circ$ and quadrant IV gives $360^\circ - 30^\circ$. Step 5: $\boxed{\theta = 210^\circ}$ and $\boxed{\theta = 330^\circ}$. Step 6: check: $\sin 210^\circ = -\frac{1}{2}$ and $\sin 330^\circ = -\frac{1}{2}$ ✓

**7. Solve $2\cos x - \sqrt{2} = 0$ on $0 \le x < 2\pi$.**
Step 1: isolate: $\cos x = \frac{\sqrt{2}}{2}$, a positive value. Step 2: expect two solutions — a horizontal line meets one period of the cosine graph twice — and cosine is positive in quadrants I and IV, so that is where they lie. Step 3: reference angle: $\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$, so the reference angle is $\frac{\pi}{4}$. Step 4: quadrant I gives $x = \frac{\pi}{4}$; quadrant IV gives $x = 2\pi - \frac{\pi}{4}$. Step 5: $\boxed{x = \dfrac{\pi}{4}}$ and $\boxed{x = \dfrac{7\pi}{4}}$. Step 6: check both in the original equation: each cosine is $\frac{\sqrt{2}}{2}$, and $2 \cdot \frac{\sqrt{2}}{2} - \sqrt{2} = 0$ ✓

### 8. Challenge: a student writes $\cos 3 = 0.9986$.

*Parts (a) and (c) are open explanations — flagged for manual review. Model answers below; part (b) is machine-verified.*

**(a)** A bare number can be read two ways: 3 radians, which is just under $\pi \approx 3.14$ and therefore nearly a half turn, or $3^\circ$, a very small angle near the positive $x$-axis. The student assumed degrees — 0.9986 is $\cos 3^\circ$, a cosine barely below its maximum, which only a nearly-zero angle can produce. By convention, an angle written with no degree symbol is in radians.

**(b)** Step 1: place 3 radians on the circle: $\frac{\pi}{2} \approx 1.57$ and $\pi \approx 3.14$, so 3 sits between them. Step 2: evaluate in radian mode: $\cos 3 = \boxed{-0.99}$ (to two decimal places).

**(c)** The angle lies in **quadrant II**, where the $x$-coordinate is negative; cosine reads that $x$-coordinate, so the answer had to be negative before any button was pressed. Being just short of $\pi$, the point is close to $(-1, 0)$, so a value near $-1$ is exactly what to expect — the student's 0.9986 has the right magnitude and the wrong sign, the signature of a degree reading of a radian angle.

*Full credit on (a) and (c) needs:* both readings of the bare 3 named with the radian convention stated, quadrant II identified with a reason, and the sign of the cosine deduced from the sign of the $x$-coordinate rather than from the calculator.